

FINAL-YEAR PROJECT • PROGRESS REPORT • AIR UNIVERSITY
ISLAMABAD

Femora: A Multilingual, Multi-Domain Women’s Health Companion

Cycle prediction, PCOS and breast-cancer screening, and a generative-AI companion in one on-device application — a technical progress summary prepared for the project supervisor.

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CONTENTS

1

Introduction & objectives

2

System architecture

3

Completed modules

4

Datasets

5

Model evaluation results

6

Distinguishing contributions

7

Testing and validation rigour

8

Limitations and threats to validity

9

Remaining work

10

Conclusion

11

References

01 Introduction & objectives

Femora is a mobile application that brings three normally separate strands of women’s health — menstrual-cycle tracking, polycystic ovary syndrome (PCOS)

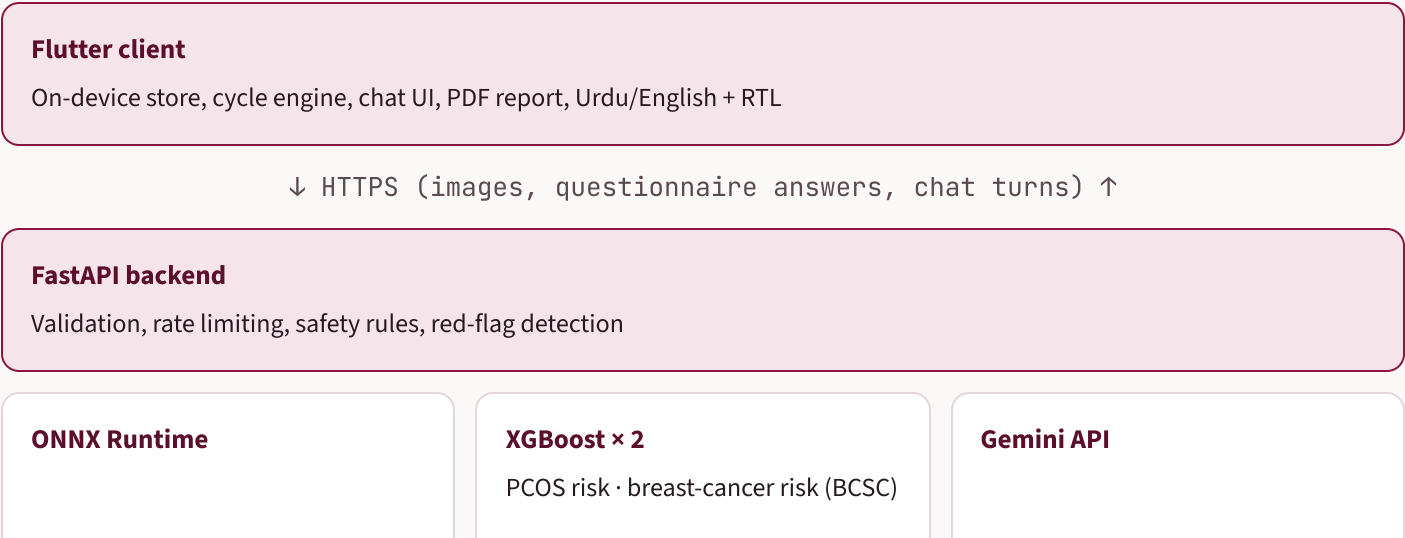
risk screening, and breast-cancer awareness — together with a generative-AI health companion, in a single on-device app aimed at women in Pakistan.

The agreed project scope defines three domains: (1) menstrual-cycle logging with next-cycle prediction; (2) a PCOS risk questionnaire scored by a gradient-boosted classifier; and (3) breast-cancer awareness through an ultrasound-image classifier, a symptom/risk questionnaire, and a chatbot that explains the results. Pregnancy care, originally domain four, was moved to a later project phase (FYP III) on the supervisor’s guidance, so the scope covered here is domains one through three plus the AI companion.

Beyond the original scope, three features were added during this reporting period to make the screening tools more usable and more accessible to their intended audience: an interactive “what-if” simulator for the PCOS estimate, a photographed-report reader that explains a lab or ultrasound report in plain language, and an Urdu/English language switch covering the whole app’s navigation and forms. These are described in Section 6.

02 System architecture

The client is a Flutter application (Android, tested; iOS untested) that stores a user’s results and logs on-device, keyed per signed-in account. A FastAPI backend hosts the three trained models and mediates every call to Google’s Gemini API^[16], so that neither the Gemini key nor the safety rules that constrain it are ever shipped inside the installable app.



ResNet50 ultrasound classifier +
non-breast gate

Chat, speech in/out, report reading
(Google, cloud)

Firestore Auth

Email, Google, phone-code and guest sign-in

Emergency and safety handling (e.g. a message describing a breast lump, or suicidal language) is detected in backend code, in English, Roman Urdu and Urdu script, and is never left to the language model’s judgement — it fires identically whether or not Gemini itself is reachable.

03 Completed modules

The table below is a compressed version of the module-status table maintained in the full engineering report (scope doument/Femora – Project Progress and Technical Report.docx, 60 pages, rebuilt automatically from the codebase and measured results).

- Menstrual-cycle tracking & prediction**

Calendar, phases, next-period / ovulation / fertile-window estimate; personal average blended with a study average, chosen over a Random Forest after direct comparison (Section 5).
- PCOS risk screening**

Questionnaire scored by an XGBoost classifier with monotonic constraints^[8] and per-answer SHAP explanations^[10].
- Interactive PCOS “what-if” simulator**

Live re-estimate as exercise, fast-food frequency or weight are changed, plus a ranked “quick wins” view; direction of every effect is verified by a 128-case monotonicity test.
- Breast-ultrasound classifier**

ResNet50^[7] convolutional network with a learned gate that rejects non-breast ultrasounds, trained on three combined public datasets and evaluated on an unseen fourth.
- Breast-cancer risk questionnaire**

XGBoost model trained on 2.39 million real screening mammograms from the Breast Cancer Surveillance Consortium^[6]; symptom-based referral follows NICE guideline NG12^[14] rather than the model.

- **Breast self-exam guide & reminder**

Step-by-step illustrated guide, logging and a monthly notification.

- **Symptom, mood & wellness tracker**

Daily symptoms, mood, sleep, stress and energy; 7/30/90-day trend charts and rule-based notes.

- **Hormonal-health insights**

Logged symptoms and mood analysed across the four cycle phases, with a typical-hormone-curve chart and recurring-symptom flags.

- **Reminders**

Period, fertile-window, ovulation, daily-log, medication and self-exam notifications, replanned whenever a period is logged.

- **Health dashboard & PDF report**

Cycle history, a backtest of the prediction against the user's own logged cycles, symptom patterns, and a one-tap lab-style PDF with charts, reference ranges and a QR code.

- **Generative-AI health companion**

Gemini-based chat with Urdu/English speech in and out, a name-free personal context drawn from the on-device store, and server-side safety rules independent of the model.

- **Photographed-report reader**

Photographs a lab, hormone or ultrasound report (or a prescription) and returns a structured, plain-language explanation in the user's chosen language; identifiers are scrubbed and a critical value always forces an urgent flag, regardless of what the model returned.

- **Urdu / English language switch**

Every screen's navigation, forms, dialogues and buttons are bilingual, with automatic right-to-left layout driven by Flutter's own locale mechanism and a bundled Nastaliq Urdu font^[18]. Generated clinical explanations and the PDF report remain English-only pending a native-speaker review (Section 8).

- **Onboarding & accounts**

First-launch profile plus email, Google, phone-code or guest sign-in (Firebase Authentication); results are kept on the phone, separately per account.

- **Pregnancy care**

Moved to project phase III on the supervisor's guidance; not started.

04 Datasets

Three screening tasks each draw on a distinct, cited dataset; no personally identifying data is collected by the app itself.

DATASETS USED FOR TRAINING AND EVALUATION			
TASK	DATASET	SIZE	CITATION
Cycle-length prediction	Fehring fertility-awareness charting data, Marquette University	159 women, 1,665 cycles	[13]
PCOS risk	PCOS dataset (clinical & lab features, 10 hospitals, Kerala, India)	541 women	[12]
Breast-ultrasound classification (training)	BUSI	766 images, 600 women	[1]
	BrEaST-Lesions-USG	256 scans, 256 patients	[2]
	BUS-BRA (adds a fourth scanner / hospital)	1,875 images, 1,064 patients	[3]
	BUS-UCLM (tried, not deployed — Section 5)	683 images, 38 patients	[4]
Held-out ultrasound test set	BUSI_WHU, Renmin Hospital of Wuhan University	927 scans	[5]
Breast-cancer risk	Breast Cancer Surveillance Consortium Risk Estimation Dataset	2.39 million mammograms	[6]
Reference-model comparison only	Breast Cancer Wisconsin (Diagnostic)	569 cases	[15]

Population caveat. The Fehring cycle data, the PCOS dataset and the BCSC mammography data were all collected outside Pakistan (the U.S. and India respectively); none has yet been validated on a Pakistani cohort, and this is recorded as a limitation of every screening estimate the app currently produces (Section 8).

05 Model evaluation results

All figures below are measured on data held out from training, never used to choose the reported operating point unless stated otherwise.

84.4%

PCOS accuracy ·
AUC 0.88 (109-woman
test split)

71.8%

Ultrasound accuracy at
the screening threshold
(710 scans)

89.2%

Ultrasound sensitivity
(cancers caught) at that
threshold

0.87

Ultrasound AUC on
scans from a hospital
never seen in training

0.64

Breast-cancer-risk AUC
(BCSC); calibration 1.01

**~67% /
82%**

Next period within 3
days — day one / after
3+ logged cycles

Breast-ultrasound classifier: generalisation was the central finding

An early version of the classifier, trained only on BUSI and BrEaST-Lesions-USG, scored 79.7% accuracy on its own held-out split but generalised poorly: on scans from a hospital it had never seen during training it caught only 72% of cancers (AUC 0.69). Adding a third dataset from a different hospital and scanner population, BUS-BRA^[3], with a strictly patient-level train/test split, raised the AUC on unseen patients from 0.69 to 0.87 and cancers caught from 72% to 91% at the notebook's own threshold. This is treated in the project as the central empirical result of the breast-imaging module: single-hospital training data measurably overstates real-world performance, and combining hospitals is what closed most of that gap.

A further candidate dataset, BUS-UCLM^[4], was tried under the same patient-level protocol and gave no measurable gain on the held-out set; it is documented but was not deployed — a negative result reported for completeness rather than omitted.

The deployed model exposes one classifier at two decision thresholds: a screening threshold (0.25) chosen to trade a small loss of sensitivity for materially fewer false alarms (accuracy 71.8%, sensitivity 89.2%, specificity 63.6%), and a neutral

threshold (accuracy 80.4%, sensitivity 77.5%, specificity 82.6%); the underlying ranking (AUC 0.87) is identical at either threshold. The screening threshold was selected after inspecting the test set and is reported as such, not as an independently confirmed operating point.

PCOS risk classifier: model choice was cross-validated, not assumed

A 5-fold stratified cross-validation compared logistic regression, a 300-tree random forest, and XGBoost^[8] on the self-reportable feature set; the three were statistically indistinguishable (AUC 0.87–0.88). XGBoost was kept because it is named in the project scope, supports monotonic constraints (a symptom can only ever push the estimate in a clinically consistent direction, never the reverse), and yields per-answer SHAP^[10] attributions used to generate the “main factors” shown to the user. Age was deliberately excluded from the model after it was found to encode a spurious rule from the narrow age range of the training cohort.

Breast-cancer risk questionnaire: calibration over discrimination

Trained on the BCSC Risk Estimation Dataset^[6], the model reaches AUC 0.64, in line with published risk-factor models of this kind, whose value lies less in ranking individuals than in calibration: the ratio of expected to observed cancers on the held-out split is 1.01. Symptom-based urgency (a lump, nipple discharge, a skin change) is handled separately by fixed referral rules drawn from NICE guideline NG12^[14], since no public outcome dataset links self-reported symptoms to diagnosis in a way a questionnaire could learn from.

Cycle-length prediction: a simple estimator, chosen by measurement

On the Fehring dataset^[13], blending a woman’s own average cycle length with the study population average predicts the next period within 3 days about 67% of the time from a single logged cycle, rising to about 82% once three or more cycles are logged. A random-forest model incorporating age and BMI was trained and compared directly; it did not improve on the simpler blended estimator, so the simpler model was deployed — a deliberate, measured preference for simplicity over model complexity where complexity bought nothing.

06 Distinguishing contributions

Beyond implementing the three scoped screening domains, three features were designed and built to close the gap between a working model and a tool a woman would actually use and trust:

An interactive, direction-verified “what-if” simulator

Rather than presenting the PCOS estimate as a single static number, the app lets a user move sliders for exercise, fast-food frequency and weight and see the model’s estimate update live, alongside a ranked list of single changes (“quick wins”). Because the underlying XGBoost model is trained with monotonic constraints, every direction shown is guaranteed consistent (e.g. weight loss below a BMI of 18.5 is never offered as a suggestion); this guarantee is checked by an automated test that runs 128 distinct answer combinations through the endpoint and asserts that exercise, less fast food and weight loss never raise the estimate, individually or combined.

A photographed medical-report reader with safety rules enforced in code, not only in the prompt

A woman can photograph a lab test, hormone panel, ultrasound report or prescription and receive a structured explanation of what it shows. The safety properties that matter clinically — never inventing a value not printed on the report, never suggesting a medicine or dose, always escalating urgency when a value is marked critical, and removing identifying information such as CNIC numbers and phone numbers from the response — are enforced by server-side code that post-processes the model’s output, not left to the prompt alone, and each is covered by an automated test.

An Urdu/English switch built on the platform’s own accessibility mechanism

Rather than hand-building right-to-left layout, the whole app’s locale is driven through Flutter’s standard localisation mechanism, so right-to-left mirroring, and Urdu translations of Flutter’s own built-in dialogs, are inherited directly from the platform rather than re-implemented. A bundled Noto Nastaliq Urdu font^[18] is attached only as a text-style *fallback*, so every existing screen automatically renders

Urdu script in a proper Nastaliq face without each of the hundreds of existing text widgets in the codebase needing to be touched individually. Every screen's own navigation, forms, dialogues, buttons and error messages are now bilingual; the generated clinical explanations and the PDF report are deliberately left in English pending a native Urdu speaker's review, discussed in Section 8.

07 Testing and validation rigour

The project is developed test-first at the level of individual behaviours: every model-facing endpoint, screen and piece of user-visible logic is covered by an automated test before being considered complete, and the count grows with the feature set rather than being added retrospectively.

350

Flutter (app) automated tests, all passing

97

Backend (Python) automated tests, all passing

0

flutter analyze errors or warnings

Testing spans unit tests of pure functions (the cycle engine, the hormonal-insight rules, the reminder planner), widget tests that render whole screens and simulate taps and text entry, and backend tests that exercise every endpoint against faked and real Gemini responses, including deliberately malformed and adversarial inputs (a prompt-injection attempt embedded in a photographed document, for instance). Model claims are treated the same way as software claims: every reported accuracy, AUC or calibration figure is computed by a script that reads the model's own saved metadata, so the number in this report and the number the code actually produces cannot drift apart silently.

08 Limitations and threats to validity

- **Population mismatch.** The cycle-prediction, PCOS and breast-cancer-risk models are all trained on data collected outside Pakistan (the Fehring^[13] cohort, a Kerala hospital cohort^[12], and the U.S. BCSC cohort^[6] respectively). None has been validated on Pakistani women, and the reported accuracy

figures should be read as evidence the methodology works on the data available, not as a claim of accuracy for the app's intended users.

- **Generative-AI privacy and correctness.** The companion and the report reader depend on Google's Gemini API^[16]. On the free tier, submitted text may be used by Google to improve its products; a paid key with a data-use opt-out is a prerequisite before the app is given to real users. The model can also misread a photographed value, which is why every reader result carries a disclaimer and asks the user to check it against the original document.
- **Urdu content coverage.** The language switch currently covers every screen's own navigation, forms and dialogues, but the generated clinical explanations (hormonal-insight flags and patterns, PCOS/breast-cancer guidance text, the text of an actual phone reminder notification, the self-exam guide's own instructions) and the PDF report remain English-only. These were deliberately left rather than translated quickly, since they carry the most clinical weight and would benefit most from review by a native Urdu speaker, which has not yet happened for any Urdu text in the app.
- **Device validation.** Several modules built in the most recent working period have not yet been exercised on a physical phone, only in the automated test suite and, for a subset, in a debug build; a release APK build and an on-device pass are recorded as outstanding.
- **Scope not yet started.** Pregnancy care, originally the project's fourth domain, has not been started and is planned for project phase III.

09 Remaining work

1. Native-speaker review and correction of all Urdu text in the app, followed by translation of the remaining generated clinical content and the PDF report.
2. Improve the companion's answer quality and move its synthesised voice closer to natural speech, which requires either a paid text-to-speech service or a paid Gemini key (the free tier allows roughly ten spoken replies per model per day).
3. Two-way, low-latency live voice conversation with the companion.

4. AI-assisted outlining of a lesion's boundary on an ultrasound image, kept as a separate feature from the existing classifier.
5. Pregnancy care (project phase III), on the supervisor's guidance.
6. On-device validation of every feature built in the most recent working period, and a fresh release APK build.

10 Conclusion

An estimated 90% of the agreed project scope is implemented, tested and measured, with pregnancy care the principal remaining domain and deliberately deferred to project phase III. Every model-performance figure in this report is computed directly from held-out data by scripts checked into the repository, every dataset is cited to its source, and every design choice that could have gone another way — XGBoost over a marginally-tied alternative, a simple blended cycle estimator over a random forest, a screening threshold traded against a neutral one — is recorded together with the measurement that motivated it. The three additions beyond scope (the interactive what-if simulator, the photographed-report reader, and the Urdu/English switch) were built to address a specific, named gap between a working screening model and a tool a woman in Pakistan would actually trust and be able to use, and each carries its own automated test coverage and its own explicitly stated limitations.

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- [18] **Noto Nastaliq Urdu** typeface. Google Fonts. SIL Open Font License 1.1.

Femora · FYP progress report · 22 September 2026

Figures reproduced from `scripts/build_report.py`, the project's own measured-results report generator

12 Update: 25 September 2026

Nine items were added after this report was written, each with its own commit, automatic tests and, where a model or outside service is involved, a measured result. The engineering report (chapter 17) has the full detail.

0.86

Lesion outline: mean Dice on 688 lesions never seen in training (median 0.92)

100% → 0%

Companion answers ending with “see a doctor” on ordinary questions

0% → 92%

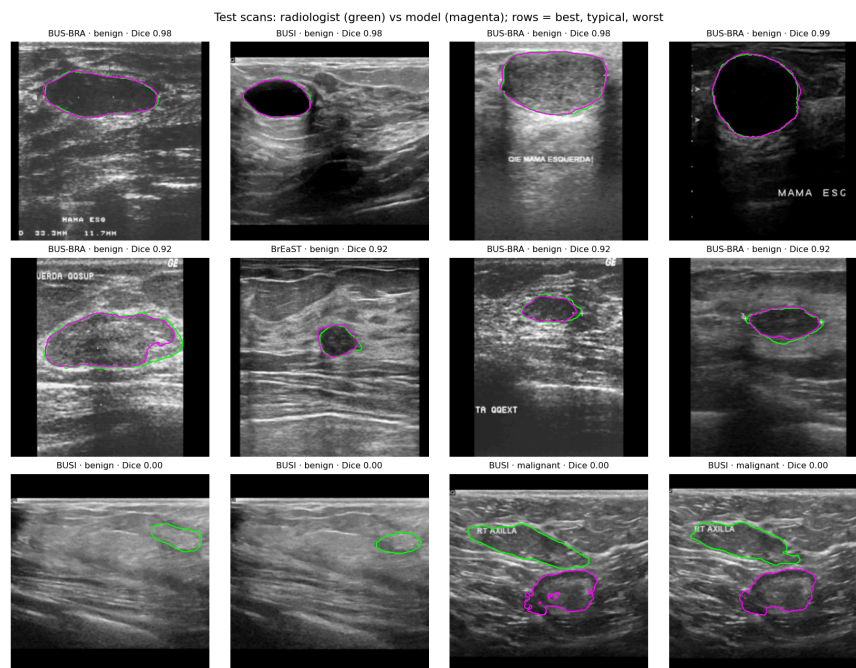
Companion answers with three or more practical steps

392 + 163

Automatic tests passing (app + server)

Lesion outline on the ultrasound

A second model, a U-Net with a ResNet34 encoder trained on Kaggle with the radiologists’ lesion masks of BUSI, BrEaST-Lesions-USG and BUS-BRA, draws the outline of the lump and gives its shape and, once the scan depth is entered, an approximate size. It uses exactly the classifier’s patient-level split, so the test lesions are unseen by both models. Mean Dice is 0.86 overall and 0.74 on BUSI, the hardest source; 95% of lesions are outlined with good overlap. Because it still draws something on a third of normal scans, the outline is shown only when the classifier has found a lesion.



Held-out test scans: radiologist’s outline (green) against the model’s (magenta); rows are the best, typical and worst cases.

A smarter AI companion

The companion now explains what is most likely going on and why, gives concrete home care and precautions, and keeps doctor advice to a specific “when to see a doctor” part. Answers are grounded in 35 short topics summarised from checked NHS, WHO and MedlinePlus pages, and the source is shown. Medicines come only from a hand-checked over-the-counter table: options unsafe for the user (pregnancy, asthma, ulcer, kidney or liver disease, blood thinners, age) are removed before the language model sees them, and it never gives doses. A trained medicine recommender was rejected because no dataset records which medicine is safe for which woman.

Same 40 questions, live Gemini, automatic scoring	Before	After
Ordinary answers ending by telling her to see a doctor	100%	0%
Answers with 3 or more practical steps	0%	92%
Topical answers based on a trusted source	0%	78%
Doses given, or ibuprofen suggested in pregnancy	0	0
Red-flag questions carrying the urgent note (two detector gaps found and fixed)	83%	100%

New features for users

- **Body clock:** an animated ring of her cycle phases with today’s position, and the typical hormone waves, on Home.
- **Show my doctor:** a QR code holding an 800-character clinical summary; nothing is uploaded, and it was checked by decoding it from a screenshot.
- **Morning heart check:** pulse from a fingertip over the camera and flash (within 3 beats per minute on synthetic recordings), linked to the cycle, with a gentle haemoglobin-test hint.
- **Femora on WhatsApp:** the same companion for texts, voice notes and report photos, ready for Meta’s free test number.
- **Nearby care:** hospitals, gynaecologists, clinics, labs and imaging centres on a map with call and directions (OpenStreetMap now; Google ratings once a key is added), linked from a suspicious scan, a high PCOS result and an urgent companion reply.
- **Profile panel and UI review:** account options and sign-out in one place; every screen checked in English and Urdu, and the wrong or untranslated elements fixed.

Not yet verified. None of these has been tried on a real phone; WhatsApp has not been connected to a real number; the companion’s medical content needs a clinician’s review and the new Urdu wording a native speaker’s. Pregnancy care remains for FYP III.

13 Update: 25 September 2026 (evening)

Four more items were added the same evening, after the app was tried in Chrome. The engineering report (chapter 18) has the full detail.

16 s → 2 s

Time to the first spoken word of an answer (natural voice, English and Urdu)

857

Characters for a full patient report inside the doctor's QR code

82

Islamabad doctors from Oladoc across four specialties, with experience and fees

174 + 399

Automatic tests passing (server + app); one older app test still failing

A natural voice that starts in about two seconds

Answers read out automatically used the phone's own robotic voice, and the natural Gemini voice took 5 to 16 seconds, allowed about 10 uses a day and did not play in a browser. The server now speaks with Microsoft's neural voices (English and Urdu, chosen from the text) and streams the audio: the first words arrive after 2.2 s in English and 1.7 s in Urdu, with no key and no daily limit. Gemini's voice remains the backup, because the Microsoft service is used without an official agreement.

The whole report behind the doctor's QR code

femora

AI WOMEN'S HEALTH SCREENING REPORT

Report ID: FEM-260925-K3P9QZ
Generated: 25 Sep 2026, 16:11

PATIENT NAME	AGE / SEX	REPORT DATE
Ayesha Khan	27 years / Female	25 Sep 2026
HEIGHT / WEIGHT	BMI	INTERESTS
162 cm / 68 kg	25.9 kg/m²	PCOS, Breast health

SUMMARY OF FINDINGS

- PCOS screening: 71% probability (HIGH).

- Breast ultrasound screening: Likely Benign (88% model confidence).

- Breast cancer risk: 1.66 x the age average (MEDIUM).

- Reported symptoms needing a doctor: Breast lump.

- Menstrual cycle: 4 cycles logged, average cycle 28.3 days (regular).

PANEL 1: PCOS RISK SCREENING

23 Sep 2026

TEST	RESULT	REFERENCE / INTERPRETATION	FLAG
PCOS risk probability	71 %	Low < 30 % Medium 30-60 % High >= 60 %	HIGH
Body mass index (BMI)	25.9 kg/m2	18.5 - 24.9 kg/m2	H
Main contributing answers	3 noted	Irregular cycles; Weight gain; Acne	-

Interpretation: the answers match patterns commonly seen with PCOS. This is a screening estimate from self-reported answers, not a diagnosis. A gynecologist can confirm with an ultrasound and hormone blood tests (LH, FSH, AMH, testosterone).

PANEL 2: BREAST ULTRASOUND AI SCREENING

24 Sep 2026

TEST	RESULT	REFERENCE / INTERPRETATION	FLAG
AI classification	Likely Benign	Normal / Benign / Suspicious (malignant)	BENIGN
Model confidence	88 %	Calibrated probability of the predicted class	-
Class probabilities	N 5 %	Benign 88 % Malignant 7 %	-
Flagging rule	>= 25 %	Flagged suspicious when the malignant probability is 25 % or more	-
Model accuracy (held-out scans)	72 %	About 89 % of cancers caught; benign scans are often flagged	-

Interpretation: the area found was rated as most likely benign (not cancer). The AI can miss cancers and can flag benign scans; follow your doctor's advice about follow-up imaging.

PANEL 3: BREAST CANCER RISK ASSESSMENT

24 Sep 2026

- Scanning the code opens **femora.web.app** with the patient's full report, laid out like the PDF report: patient block, summary, the five lab-style panels with charts, next steps and a clinician box. It animates in as it is scrolled and can be saved as a PDF.
- Still private.** The report is compressed inside the link, after the #, which a browser never sends to a server. The page (Firebase Hosting, free plan) only unpacks and draws it on the doctor's phone; nothing is uploaded or stored.
- A full test report is 857 characters; a very full one drops the daily charts first to stay scannable.

The report page on a phone (sample data), as the doctor sees it after scanning.

A Home card that keeps moving

After its intro the cycle card no longer stops: a glint of light circles the phase ring, today's marker breathes, the hormone waves rise and fall, a spark runs along the estrogen curve to today and soft lights drift behind the card. The motion stops when the phone asks for reduced motion.

Find a doctor, with Oladoc's permission

Instead of listing hospitals (already on Google Maps), Femora now lists individual women's-health doctors with a mini profile she can judge: photo, PMDC verification, qualifications, years of experience, rating and number of reviews, fee, wait time and every clinic with its availability and a Book button. She can sort by top rated, nearest, most experienced, lowest fee or most reviewed.

Google Places could not be used without a prepaid billing account. Oladoc has no public API and its terms forbid copying its listings, so the department asked Oladoc through the Vice Chancellor's office; Oladoc granted permission on 25 September 2026 to use its data for developing and demonstrating Femora, and offered an official API key later. Until then the server reads Oladoc's listing pages gently (three pages per search, one request at a time, cached for a day); every card credits Oladoc and every booking goes through Oladoc.

Live search, Islamabad	Doctors	Example (rating, reviews, experience, fee)
Gynaecologist	30	4.9 from 607 reviews · 19 years · Rs 3,000 · 3 km
Breast surgeon	17	4.9 from 1,704 reviews · 19 years · Rs 3,500 · 4.5 km
Hormones and PCOS (endocrinologist)	21	4.9 from 3,285 reviews · 33 years · Rs 5,000 · 1.4 km
Fertility consultant	14	4.9 from 1,201 reviews · 20 years · Rs 2,500 · 1.8 km

- **Speed:** about 7 seconds for the first search of a specialty, then instant from the cache.
- **Order of sources:** Google Places if a key is ever added, else Oladoc, else OpenStreetMap (names only).
- **Limits:** written patient reviews are not on the listing pages, so the profile opens them on Oladoc; reading web pages breaks if Oladoc changes its layout, which the offered API key would remove; the permission covers development and demonstration, not public release.

Not yet verified. The new voice, the report page scanned by a second phone, the moving Home card and Find a doctor were checked in Chrome and by automatic tests, not on a real phone. One older app test (the heart-rate screen) is failing and has not been investigated yet. Pregnancy care remains for FYP III.